Instruction Tuning

План

- Мотивация
- Общий подход RLHF
- Откуда брать данные?
- PPO, GRPO, DPO

- Reasoning
- DeepSeek R1

GPT для диалоговых систем

Модель, которая умеет генерировать только продолжение текста не очень полезна.

Нам хочется, чтобы она умела вести диалог.

GPT

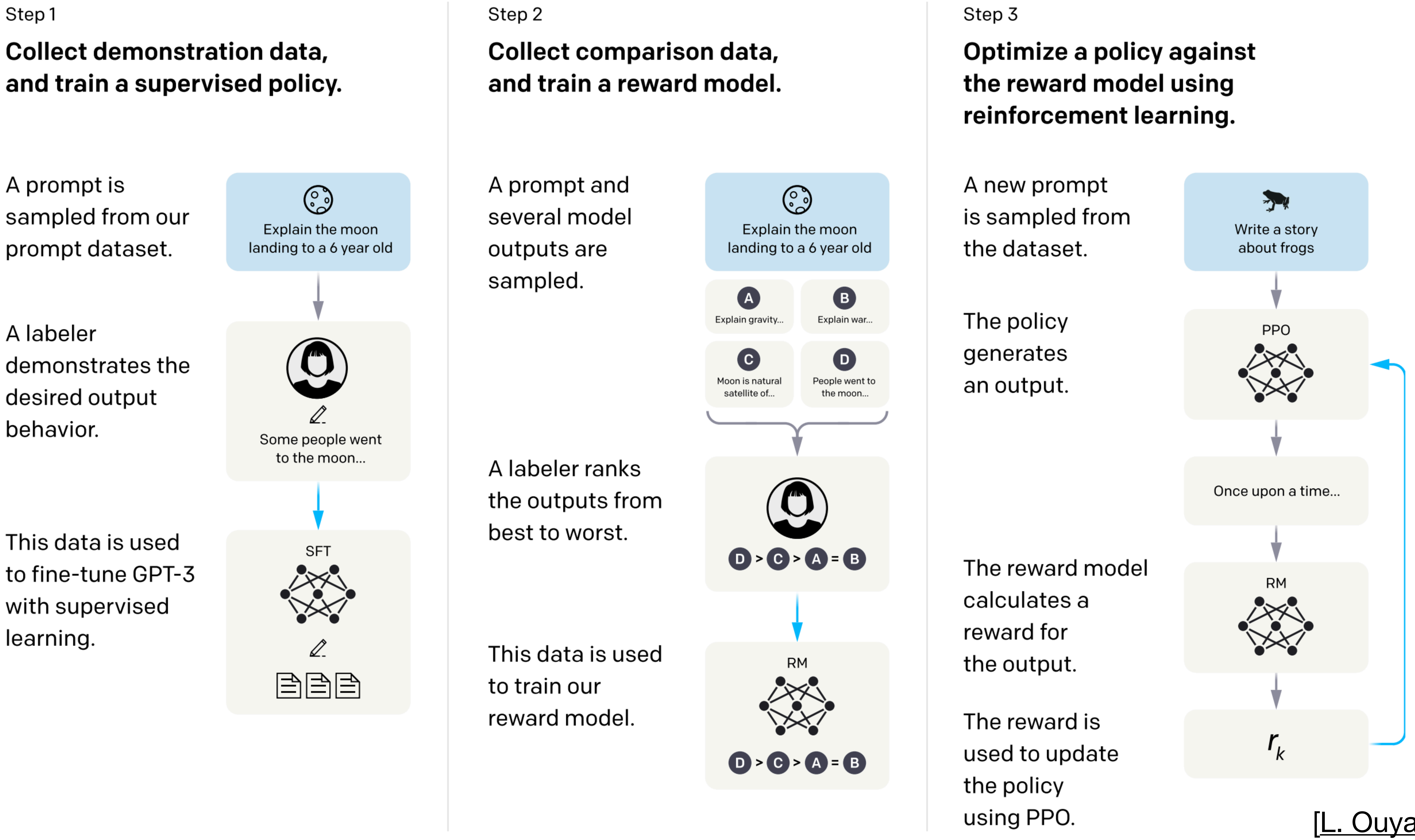
– Какая завтра погода в Москве? Какая завтра погода в Париже? Какая завтра погода в Лондоне?

ChatGPT

– "Какая завтра погода в Москве?"

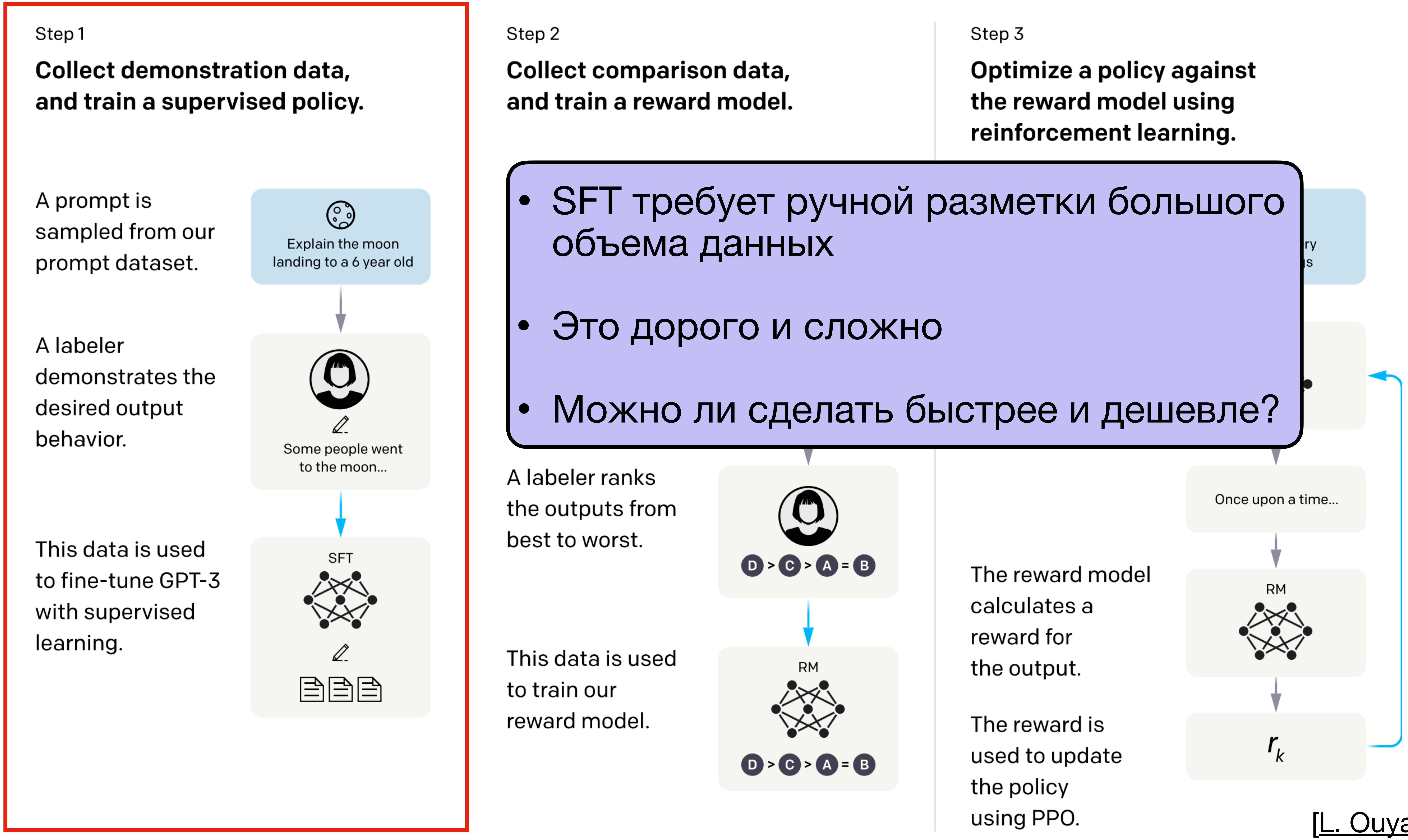
– "Завтра в Москве +15°, облачно с прояснениями."

Reinforcement Learning with Human Feedback



[L. Ouyang et al, 2022]

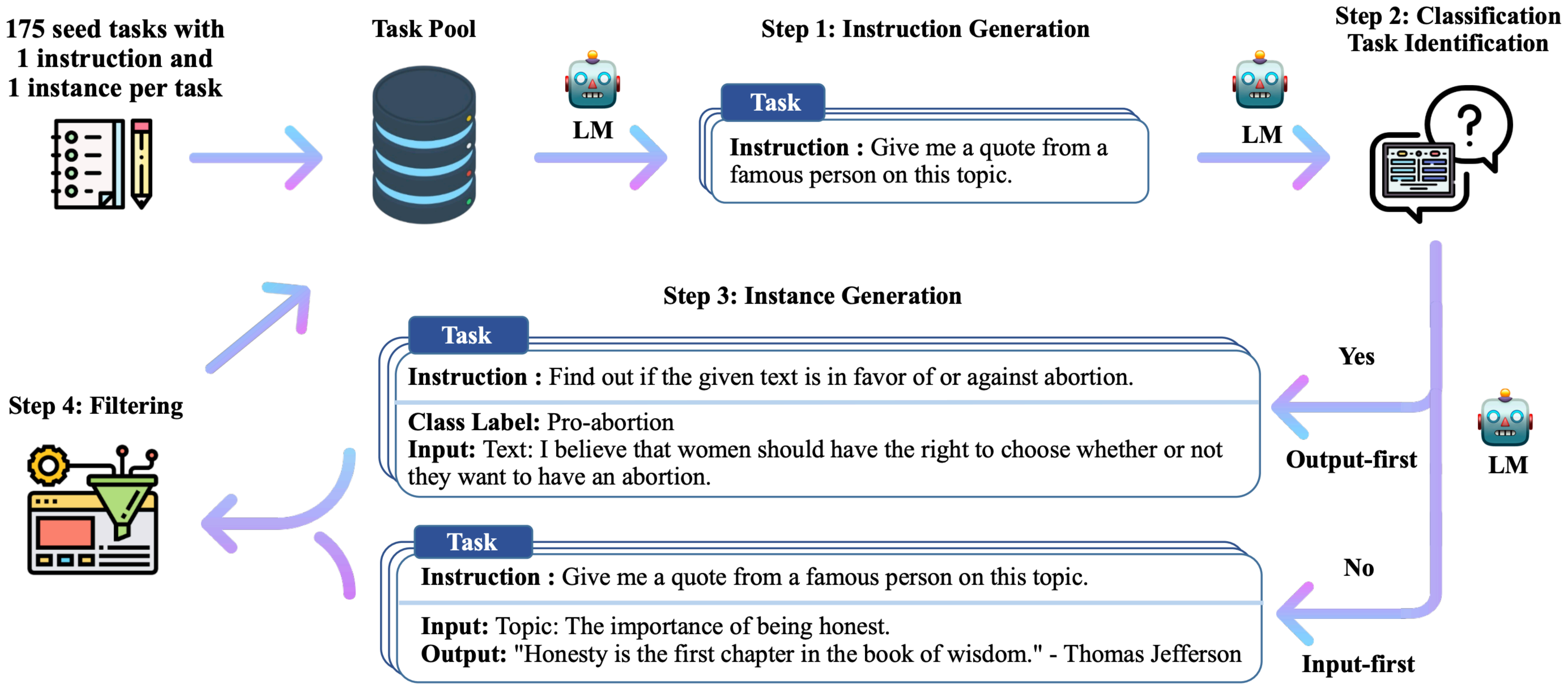
Reinforcement Learning with Human Feedback



- SFT требует ручной разметки большого объема данных
- Это дорого и сложно
- Можно ли сделать быстрее и дешевле?

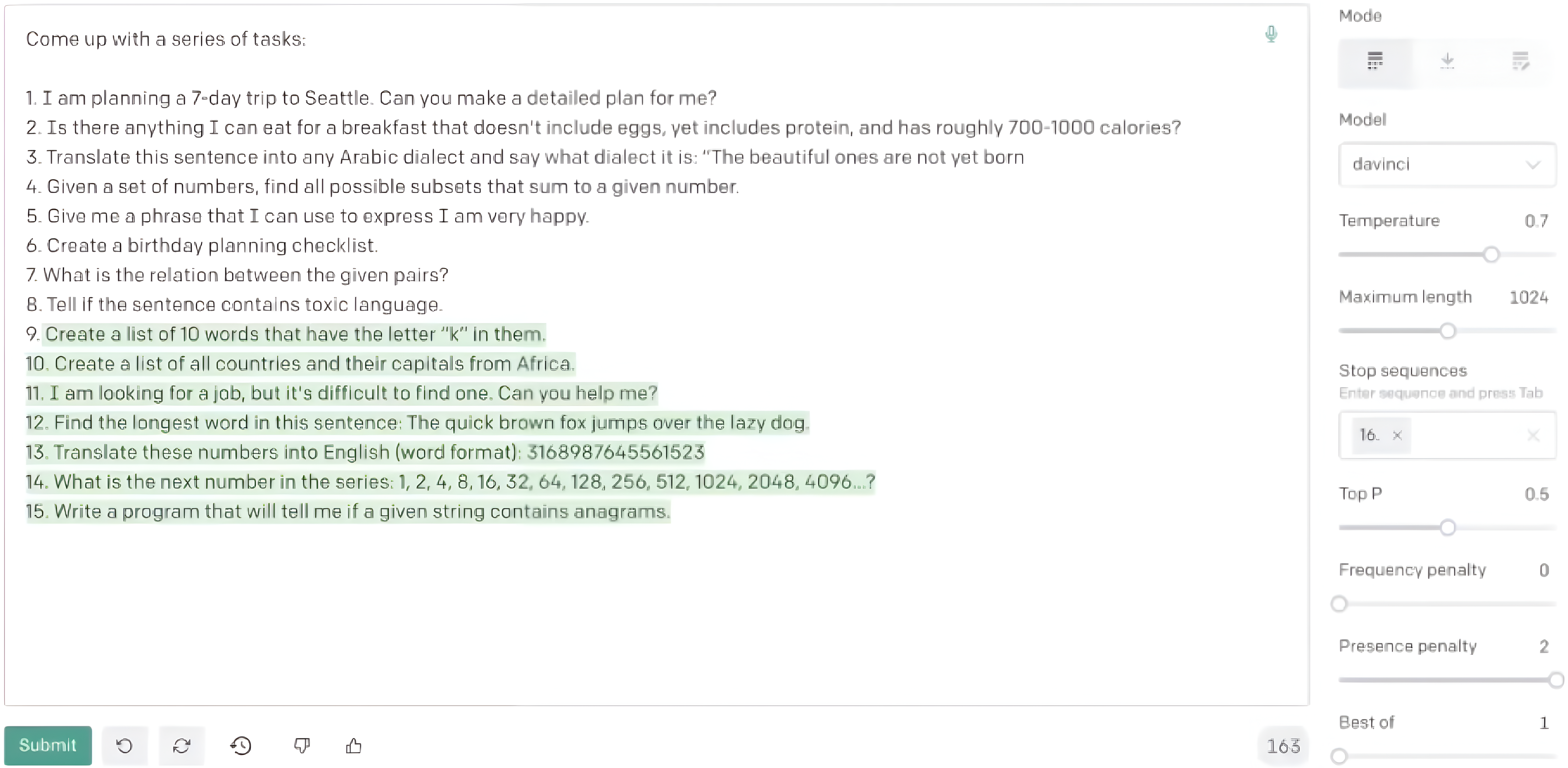
[L. Ouyang et al, 2022]

Self-Instruct



[Y. Wang et al, 2023]

Self-Instruct: генерация запросов



[Y. Wang et al, 2023]

Self-Instruct: классификация запросов

Определяем задачу: классификация или генерация

```
Can the following task be regarded as a classification task with finite output labels?

Task:  Given my personality and the job, tell me if I would be suitable.
Is it classification?  Yes

Task:  Give me an example of a time when you had to use your sense of humor.
Is it classification?  No

Task:  Replace the placeholders in the given text with appropriate named entities.
Is it classification?  No
…

Task:  To make the pairs have the same analogy, write the fourth word.
Is it classification?  No

Task:  Given a set of numbers, find all possible subsets that sum to a given number.
Is it classification?  No

Task:  {instruction for the target task}
```

[Y. Wang et al, 2023]

Self-Instruct: генерация ответов

```
Come up with examples for the following tasks.  Try to generate multiple examples when possible.
If the task doesn't require additional input, you can generate the output directly.

Task:  Which exercises are best for reducing belly fat at home?
Output:
- Lying Leg Raises
- Leg In And Out
- Plank
- Side Plank
- Sit-ups

...

Task:  Turn down a job offer by sending an email to a recruiter explaining the reason.
Output:  Hi [Recruiter],
Thank you so much for the generous offer to join your team.  As we discussed, I've admired the
company for a number of years, and am a proud endorser of its products.  However, after further
consideration of where I currently am in my career, I've decided to accept an offer at another
company.
I would love to stay in touch with you and have already started following you on [Social Media
Platform].  Again, thank you so much for your time and consideration.
Thanks again,
[Your Name]

Task:  {Instruction for the target task}
```

[Y. Wang et al, 2023]

Self-Instruct: фильтрация

Новый запрос добавляется в базу только если:

- ROUGE-L < 0.7 для любого запроса в базе
- В запросе нет специальных слов (image, graph)

Ответы фильтруются:

- По уникальности
- По статистикам (длина, повторения)

[Y. Wang et al, 2023]

Self-Instruct: результаты

Человеческая оценка качества данных

Quality Review Question	Yes %
Does the instruction describe a valid task?	92%
Is the input appropriate for the instruction?	79%
Is the output a correct and acceptable response to the instruction and input?	58%
All fields are valid	54%

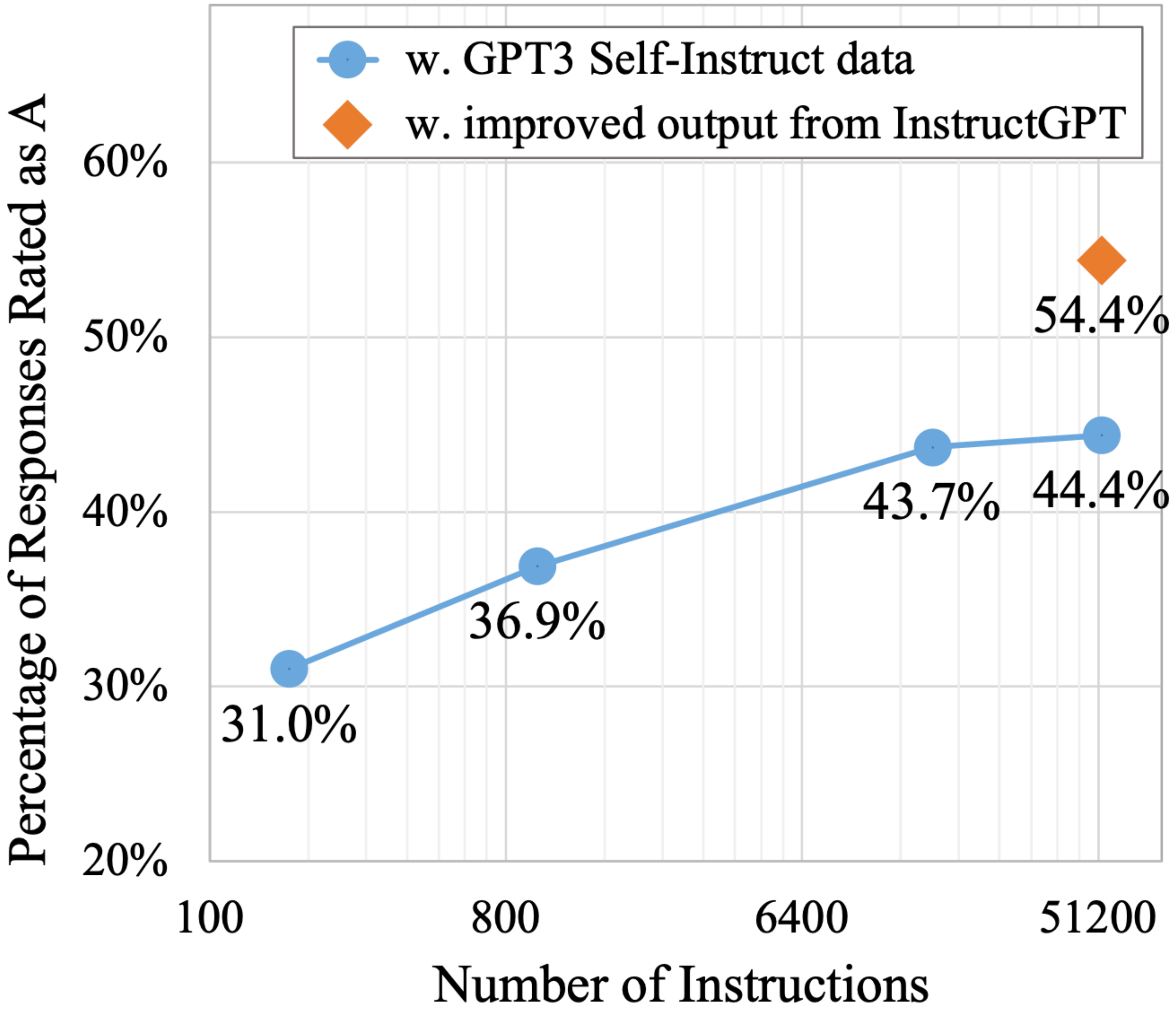
Качество на unseen примерах SuperNI

Model	**# Params**	**ROUGE-L**
Vanilla LMs		
T5-LM	11B	25.7
GPT3	175B	6.8
Instruction-tuned w/o SUPERNI		
T0	11B	33.1
GPT3 + T0 Training	175B	37.9
$\text{GPT3}_{\text{SELF-INST}}$ (Ours)	175B	39.9
InstructGPT_{001}	175B	**40.8**
Instruction-tuned w/ SUPERNI		
T*k*-INSTRUCT	11B	46.0
GPT3 + SUPERNI Training	175B	49.5
$\text{GPT3}_{\text{SELF-INST}}$ + SUPERNI Training (Ours)	175B	**51.6**

[Y. Wang et al, 2023]

Self-Instruct: результаты

Для SFT нужно не очень много данных



[Y. Wang et al, 2023]

Reinforcement Learning with Human Feedback

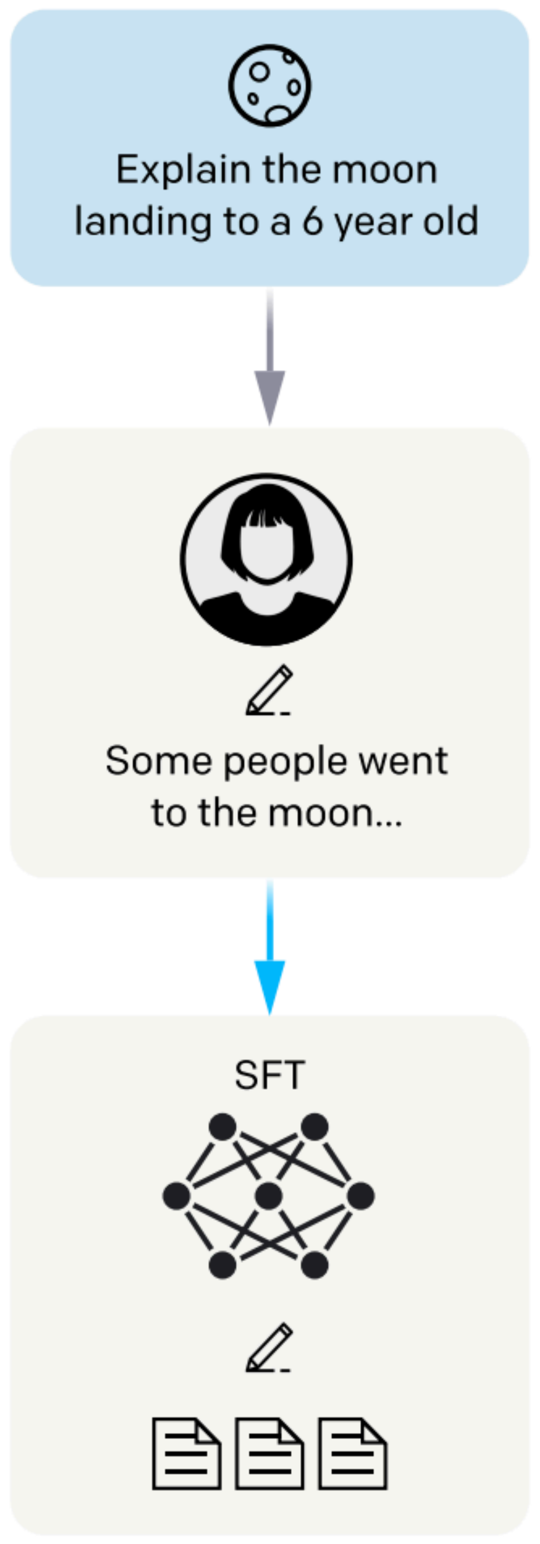
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



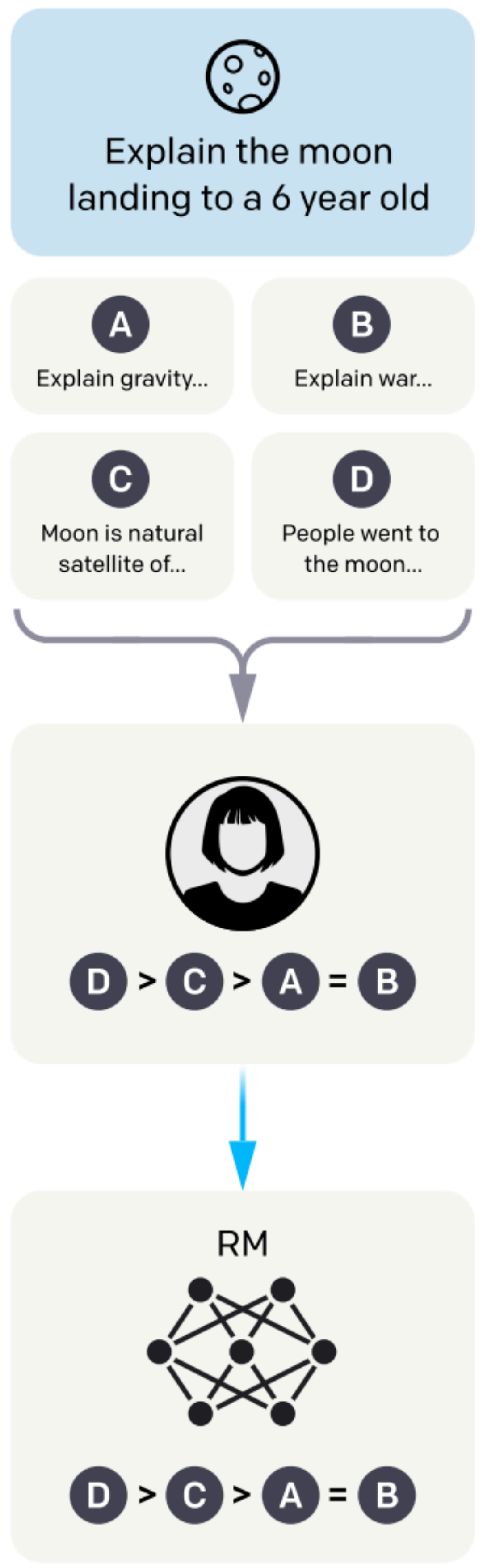
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

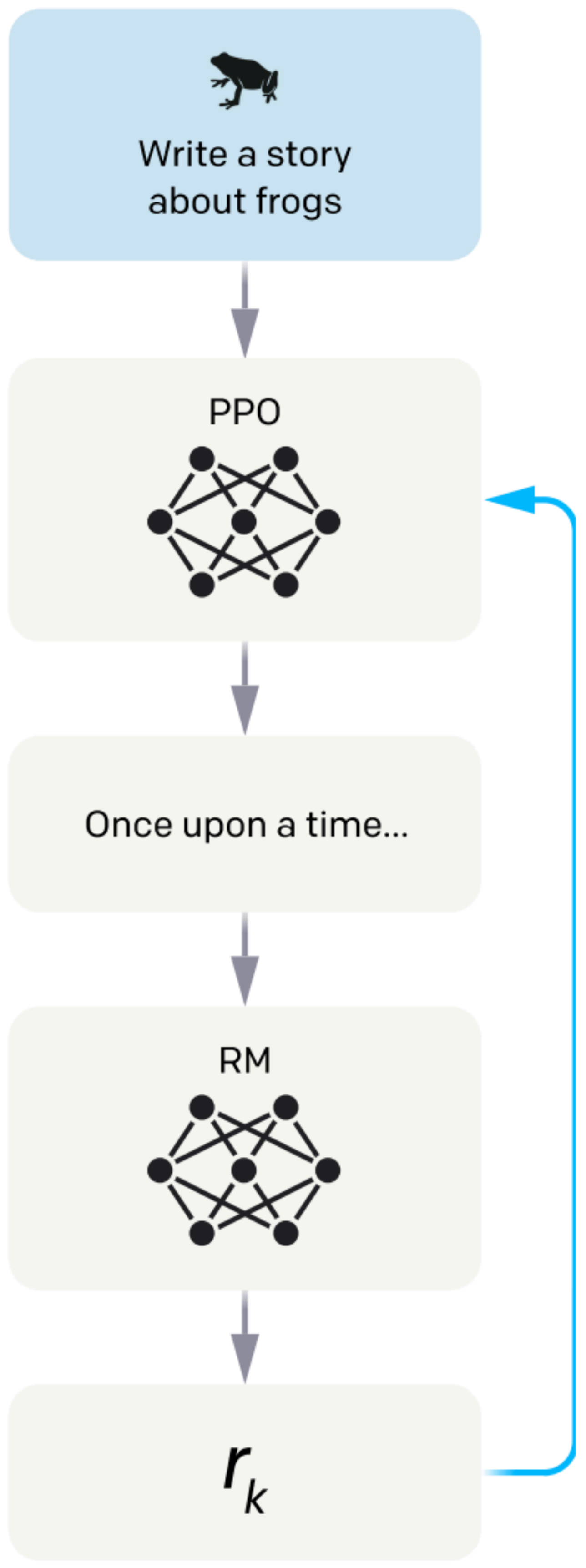
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



Reward Model

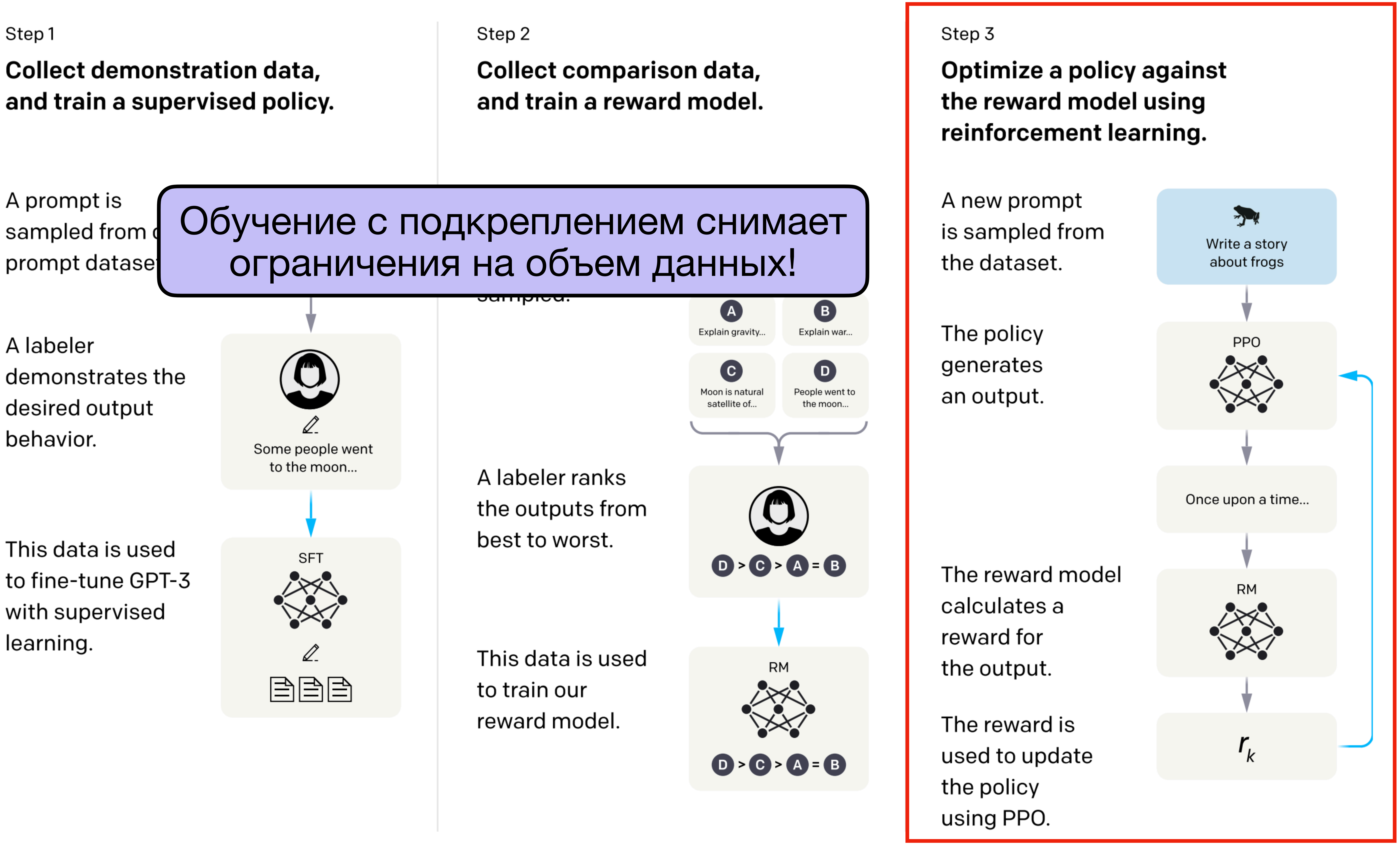
Модель учится отдавать приоритет более хорошему ответу

$$\operatorname{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} \left[\log \left(\sigma \left(r_\theta \left(x, y_w \right) - r_\theta \left(x, y_l \right) \right) \right) \right]$$

K – число отранжированных ответов ($K \in [4, 9]$)

Reinforcement Learning with Human Feedback

Обучение с подкреплением снимает ограничения на объем данных!



Trust Region Policy Optimization (TRPO)

Текущие параметры

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t\left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\hat{A}_t\right] \to \max$$

Предыдущие параметры

Advantage $\hat{A}_t = r_t(s_t, a_t) - V_\theta(s_t)$

Если $\hat{A}_t > 0$, то выбранное действие лучше других в среднем

Trust Region Policy Optimization (TRPO)

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Если $\hat{A}_t > 0$, то выбранное действие лучше других в среднем

Запрещаем сильно изменять политику

$$\text{subject to} \quad \hat{\mathbb{E}}_t[\text{KL}[\pi_{\theta_{\text{old}}}(\cdot \mid s_t), \pi_\theta(\cdot \mid s_t)]] \leq \delta$$

$$\underset{\theta}{\text{maximize}}\, \hat{\mathbb{E}}_t \left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t - \beta\, \text{KL}[\pi_{\theta_{\text{old}}}(\cdot \mid s_t), \pi_\theta(\cdot \mid s_t)] \right]$$

Proximal Policy Optimization (PPO)

$$L^{CPI}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t \right] \to \max$$

Стабилизация обучения

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min\left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t, \operatorname{clip}\left(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, 1-\epsilon, 1+\epsilon \right) \hat{A}_t \right) \right]$$

Оригинальный функционал

Защищает от слишком больших улучшений
(взламывание награды)

Proximal Policy Optimization (PPO)

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min\Big(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\,\hat{A}_t, \text{clip}\Big(\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}\,, 1-\epsilon, 1+\epsilon\Big)\hat{A}_t\Big)\Big]$$

Финальный функционал PPO

$$L_t^{CLIP+VF}(\theta) = \hat{\mathbb{E}}_t\big[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta)\big]$$

$$\boxed{\hat{A}_t = r_t(s_t, a_t) - V_\theta(s_t)}$$

Ошибка value function: $(V_\theta(s_t) - V_t^{\text{targ}})^2$

RL в RLHF

Оценка KL-дивергенции
Штраф за большие изменения

Reward Model

$$\text{objective}\,(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] +$$
$$\gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Не дает забыть общую информацию

RL в RLHF

$$\text{objective}\,(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\mathrm{RL}}}}\left[r_\theta(x,y) - \beta \log\left(\pi_\phi^{\mathrm{RL}}(y \mid x)/\pi^{\mathrm{SFT}}(y \mid x)\right)\right] +$$
$$\gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\mathrm{RL}}(x))\right]$$

Обучение **value function**

$$L^{VF}(\xi) = \mathbb{E}_{(x,y)\sim D\pi_{\phi_t}^{\mathrm{RL}}}\left[\left(r_\theta(x,y) - V_\xi(x)\right)^2\right]$$

Advantage $A(x, y) = r_\theta(x,y) - V_\xi(x)$

PPO для RL в RLHF

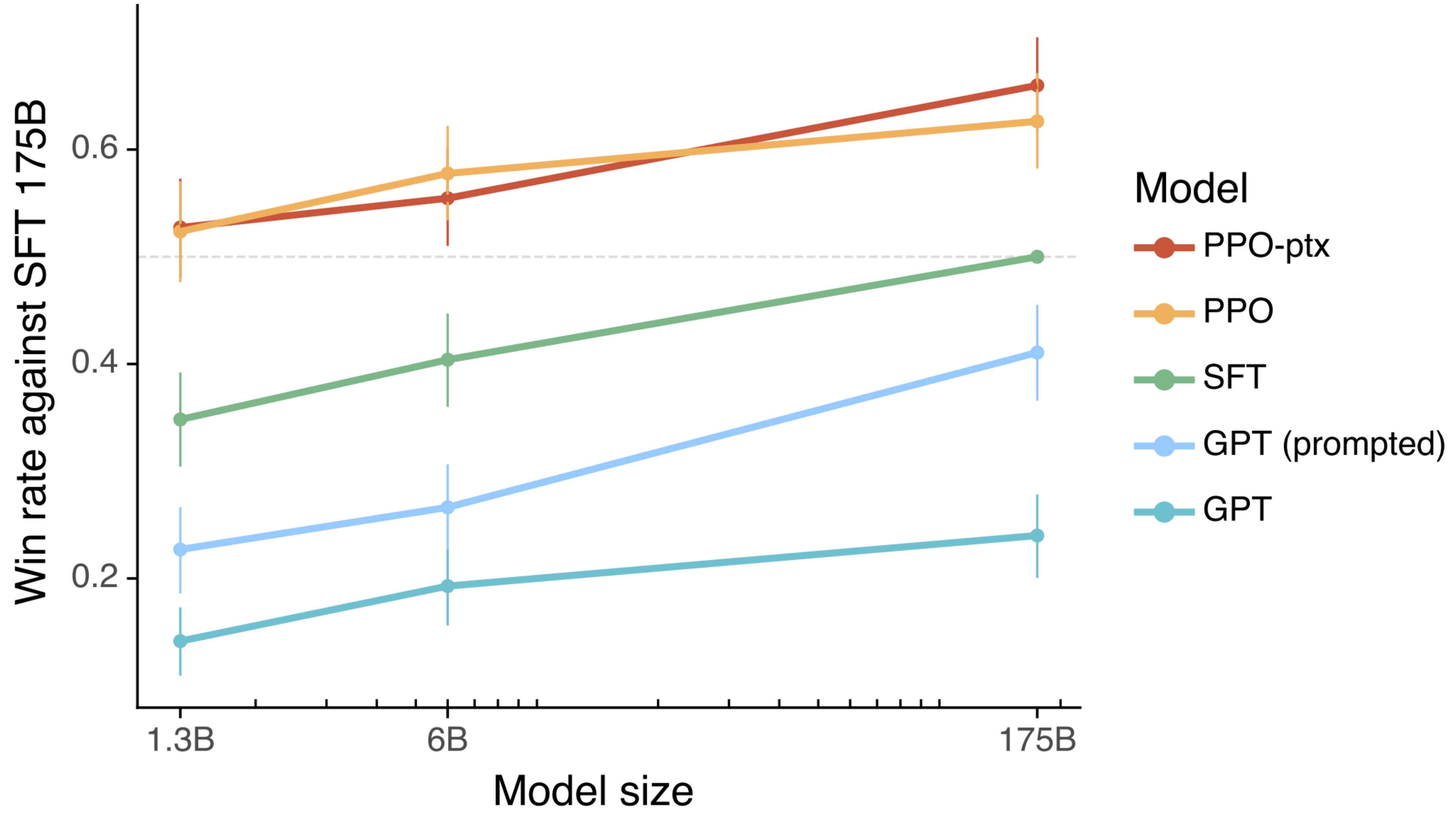
$$L_{\text{PPO-ptx}}(\phi) := E_{(x,y)\sim D_{\pi_{\phi_t}^{\text{RL}}}}\left[\min\left(\frac{\pi_\phi^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}A(x,y), \text{clip}\left(\frac{\pi_\phi^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}, 1-\epsilon, 1+\epsilon\right)A(x,y)\right)\right.$$

$$\left. - \beta\log\left(\frac{\pi_\phi^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)}\right)\right] + \gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\text{RL}}(x))\right]$$

Результаты RLHF

Table 6: Dataset sizes, in terms of number of prompts.

SFT Data			RM Data			PPO Data		
split	source	size	split	source	size	split	source	size
train	labeler	11,295	train	labeler	6,623	train	customer	31,144
train	customer	1,430	train	customer	26,584	valid	customer	16,185
valid	labeler	1,550	valid	labeler	3,488			
valid	customer	103	valid	customer	14,399			



[L. Ouyang et al, 2022]

Проблемы RLHF

- Необходимость обучения **reward model** и **value function**

Можно ли не использовать **value function**?

Group Relative Policy Optimization (GRPO)

PPO:

$$L_{\text{PPO}}(\phi) := E_{(x,y)\sim D_{\pi_{\phi_t}^{\text{RL}}}}\left[\min\left(\frac{\pi_{\phi}^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}A(x,y), \text{clip}\left(\frac{\pi_{\phi}^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}, 1-\epsilon, 1+\epsilon\right)A(x,y)\right) - \beta\boxed{\log\left(\frac{\pi_{\phi}^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)}\right)}\right]$$

Монте-Карло оценка KL-дивергенции
(Большая дисперсия)

Group Relative Policy Optimization (GRPO)

PPO:

$$L_{\mathrm{PPO}}(\phi) := E_{(x,y)\sim D_{\pi_{\phi_t}^{\mathrm{RL}}}}\left[\min\left(\frac{\pi_{\phi}^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}A(x,y), \mathrm{clip}\left(\frac{\pi_{\phi}^{RL}(y|x)}{\pi_{\phi_t}^{RL}(y|x)}, 1-\epsilon, 1+\epsilon\right)A(x,y)\right) - \beta \boxed{\log\left(\frac{\pi_{\phi}^{\mathrm{RL}}(y|x)}{\pi^{\mathrm{SFT}}(y|x)}\right)}\right]$$

Монте-Карло оценка KL-дивергенции
(Большая дисперсия)

GRPO:

$$\mathcal{J}_{GRPO}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \mathrm{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right)$$

$$A_i = \frac{r_i - \mathrm{mean}(\{r_1, r_2, \cdots, r_G\})}{std(\{r_1, r_2, \cdots, r_G\})}$$

Не используем **value function**

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1$$

Дивергенция Брегмана
(Меньше дисперсия)

Еще проблемы RLHF

- Необходимость обучения **reward model** и **value function**
- Качество модели ограничивается качеством **reward model**

- **Reward model** ограничена размером выборки

Можно ли не использовать модель награды?

Direct Policy Optimization (DPO)

Модель обучается прямо на **preference** данных

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

Недостатки DPO

- DPO требует более чистых данных
- DPO может приводить к переобучению (коллапсирование)
- Чувствительность к гиперпараметрам

Model	Method	Intro.	Inter.	Comp.
GPT-Neo 2.7B	SFT	5.6%	0.8%	0.0%
Codex 12B	SFT	9.7%	0.5%	0.1%
CodeT5	CodeRL	16.4%	4.9%	2.8%
AlphaCode 1B	5@1k	14.4%	5.6%	4.6%
Code Llama 7B	Few shot	10.8%	2.0%	0.8%
	SFT	**30.0%**	**7.8%**	**2.8%**
	DPO-Iter	20.9%	3.4%	1.3%
	PPO	29.4%	7.6%	2.4%
Code Llama 13B	Few shot	23.7%	5.6%	2.1%
	SFT	33.7%	8.7%	3.6%
	DPO-Iter	33.0%	8.0%	2.8%
	PPO	**36.4%**	**11.47%**	**4.6%**
Code Llama 34B	Few shot	32.8%	8.8%	2.9%
	SFT	38.6%	10.1%	3.9%
	DPO-Iter	34.2%	9.3%	3.7%
	PPO	**44.4%**	**18.0%**	**9.1%**

[S Xu et al, 2024]

Reasoning

Chain of Thought

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

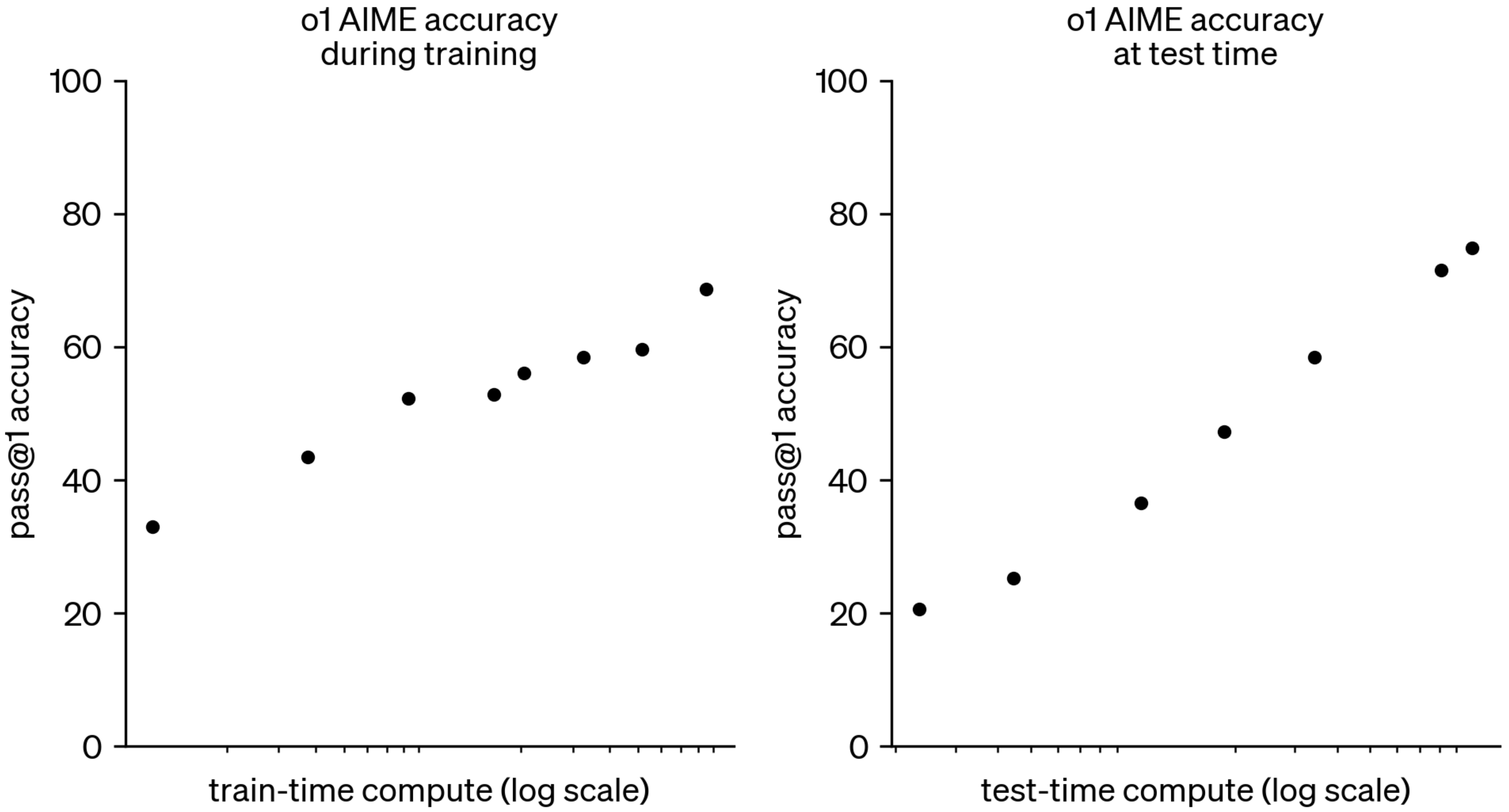
Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

[T Kojima et al, 2023]

Test-time compute scaling

Чем больше токенов генерирует модель, тем выше ее качество



[OpenAI, 2024]

Thinking tokens

В OpenAI o1 добавились "думающие" токены

```
Вопрос: Как какать не снимая свитер?

<think> ... думаю так, думаю сяк ... </think>

Ответ: В принципе, свитер не мешает
```

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Ответ: В принципе, свитер не мешает
```

Как они это сделали?

DeepSeek-R1-Zero

[DeepSeek-AI, 2025]

1) Обучение с промптом

Предобученая LLM (GPT, LLaMA) обучается генерировать ответы по промпту

A conversation between User and Assistant. The user asks a question, and the Assistant solves it.The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here </think>
<answer> answer here </answer>. User: {prompt}. Assistant:

Сложный вопрос, требующий подумать

[DeepSeek-AI, 2025]

2) Rule-based reward

Награда задается по двум правилам

1) Точность ответа

```
4. Given the three sides of an obtuse
triangle are 3, $4, x$, then the range
of values for $x$ is ( ).
(A) $1<x<7$.
(B) $5 \ll x<7$.
(C) $1<x<\sqrt{7}$.
(D) $5<x<7$ or $1<x<\sqrt{7}$.
```

Ответ $\boxed{D}$

2) Соответствие формату

```
<think>\nOkay, so I have this problem
here where I need to find the range of
values for x, given that the three sides
of an obtuse triangle are 3, 4, and x.
The answer choices are A through D. Let
me think through this step by ...
</think>
```

Проверяем, что все "думание" там, где надо

[DeepSeek-AI, 2025]

Обучение

Модель обучается с помощью GRPO, потому что
он не требует никаких доп моделей

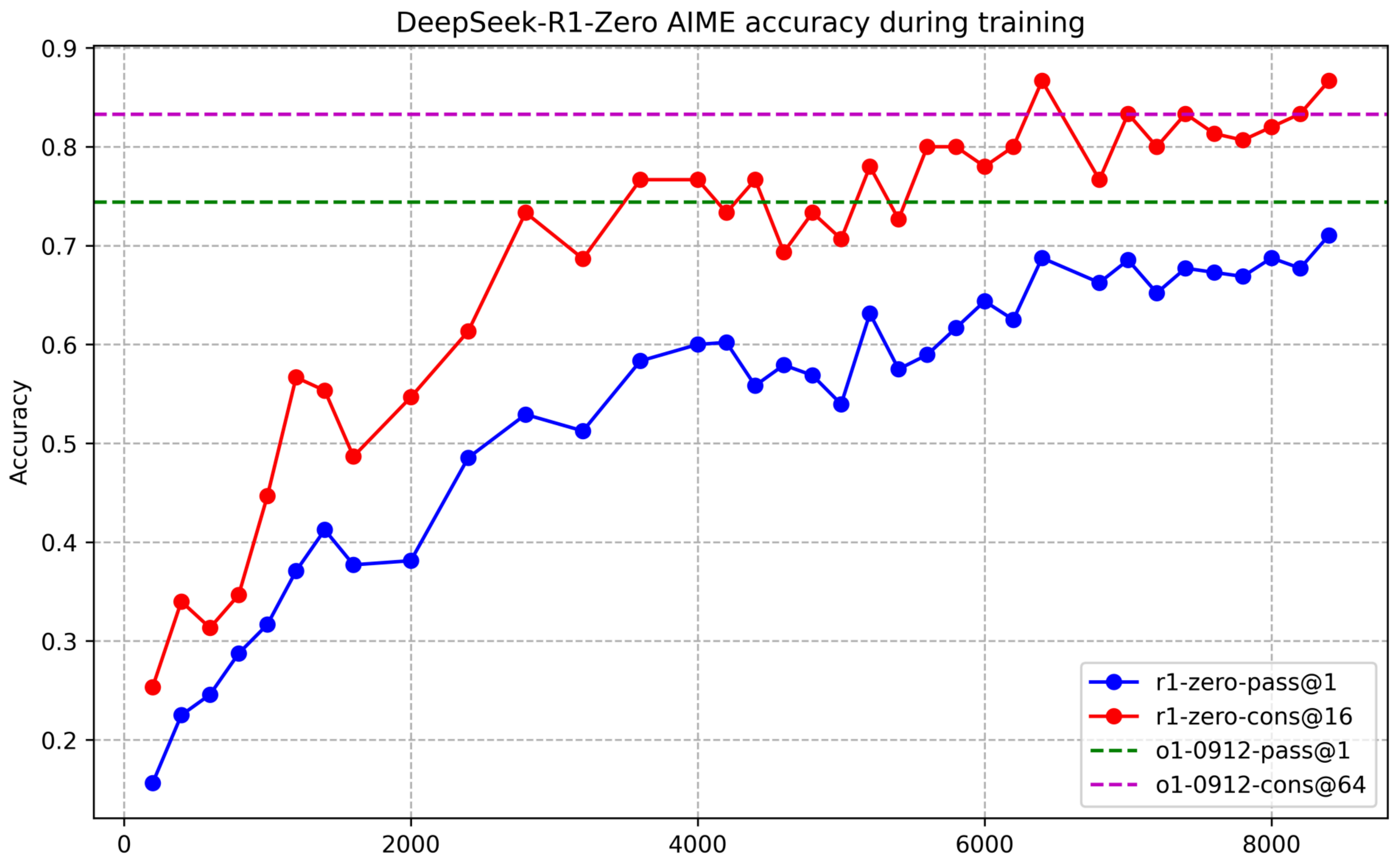
$$\mathcal{J}_{GRPO}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}A_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon\right)A_i\right) - \beta\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right)\right)$$

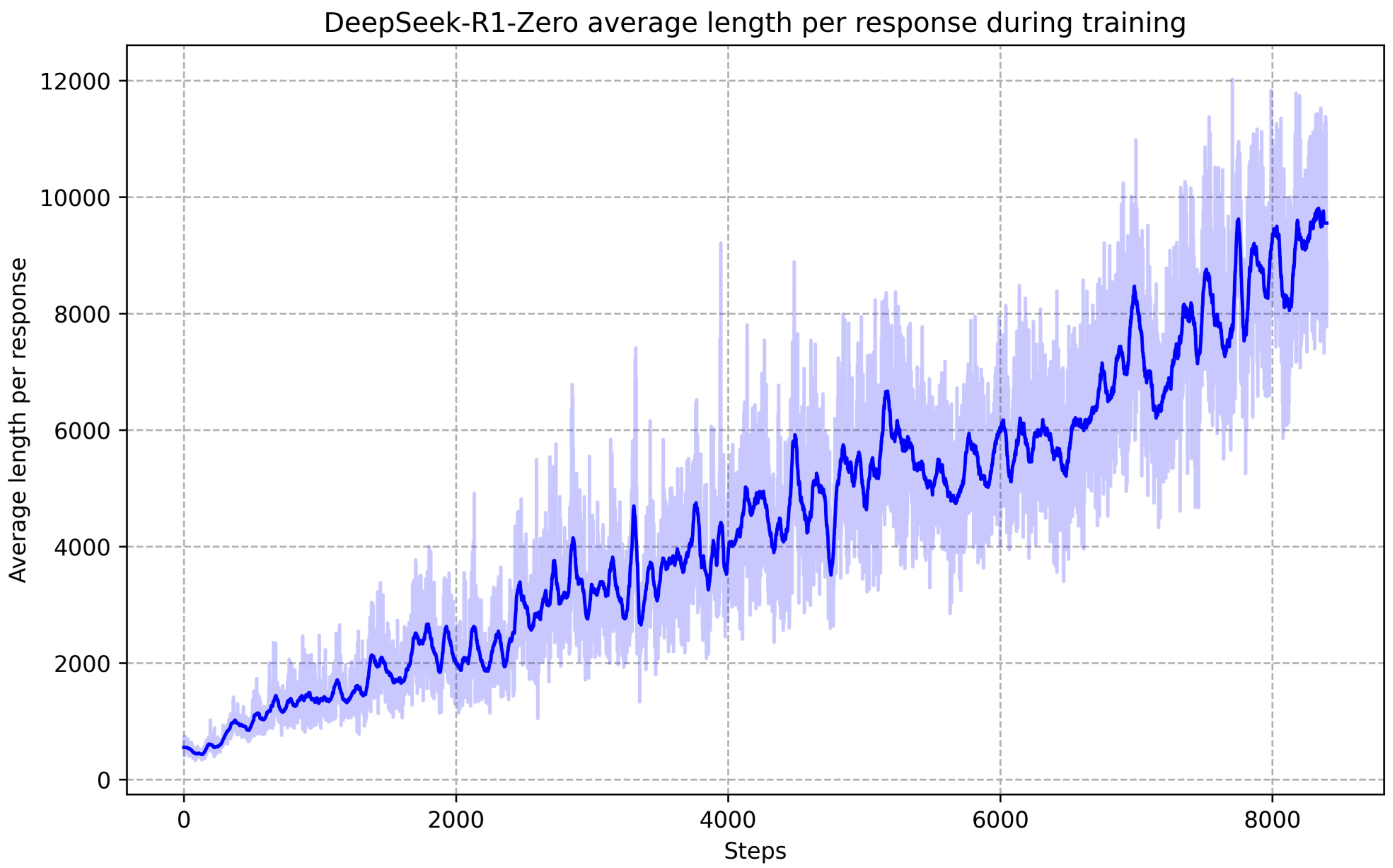
$$A_i = \frac{r_i - mean(\{r_1, r_2, \cdots, r_G\})}{std(\{r_1, r_2, \cdots, r_G\})}$$

$$\mathbb{D}_{KL}\left(\pi_\theta||\pi_{ref}\right) = \frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{ref}(o_i|q)}{\pi_\theta(o_i|q)} - 1$$

[DeepSeek-AI, 2025]

Результаты





[DeepSeek-AI, 2025]

Результаты

Модель сама научилась правильно размышлять

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a+x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a+x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a+x}}\right)^2 = x^2 \implies a - \sqrt{a+x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a+x}} = x$

First, let's square both sides:

$a - \sqrt{a+x} = x^2 \implies \sqrt{a+x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

...

[DeepSeek-AI, 2025]

Проблемы

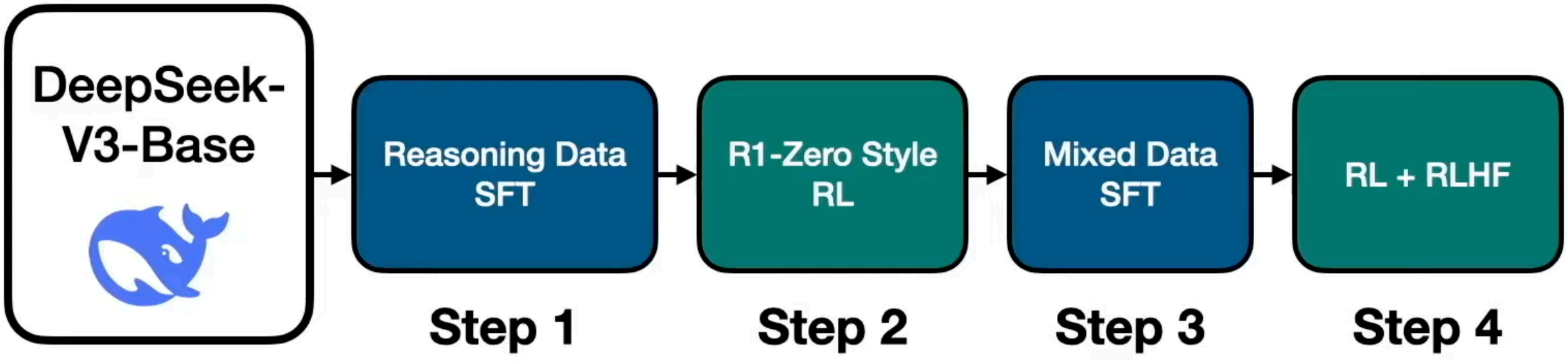
Readability

Модель не всегда отвечает в нужном формате

Language Mixing

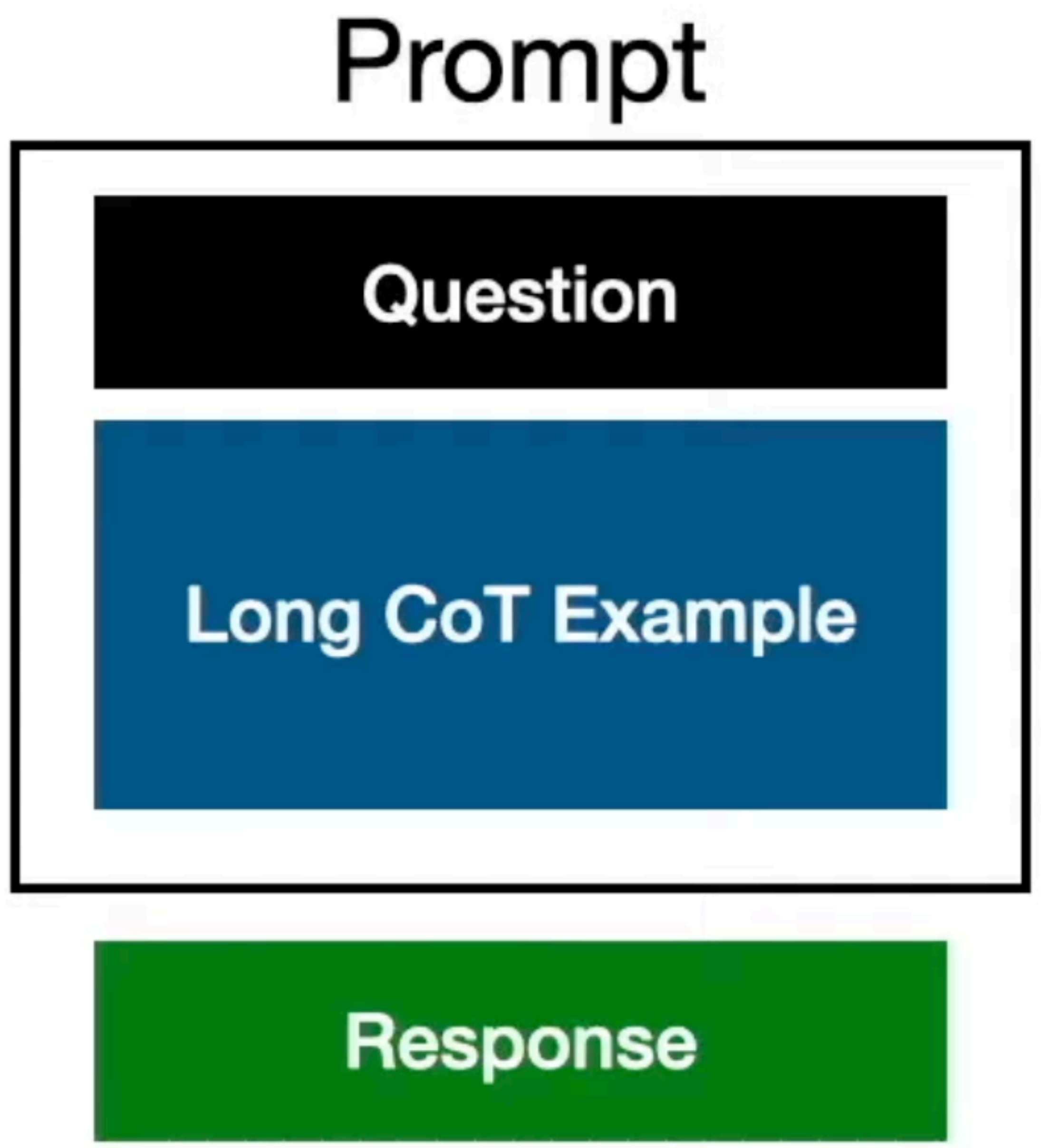
Модель отвечает не на том языке

DeepSeek-R1

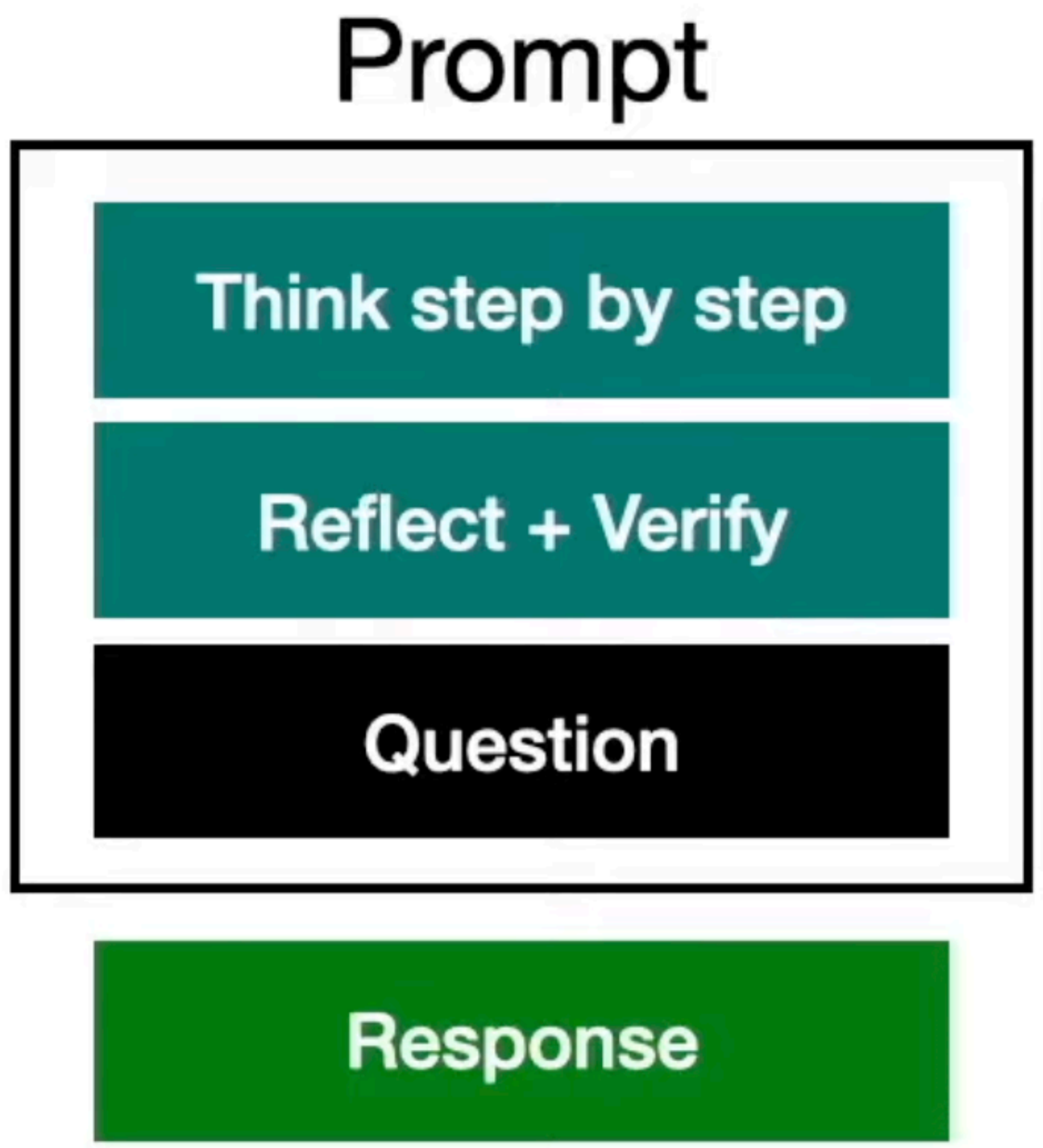

DeepSeek-V3-Base
Reasoning Data SFT
R1-Zero Style RL
Mixed Data SFT
RL + RLHF
Step 1
Step 2
Step 3
Step 4

Step-1: SFT + reasoning

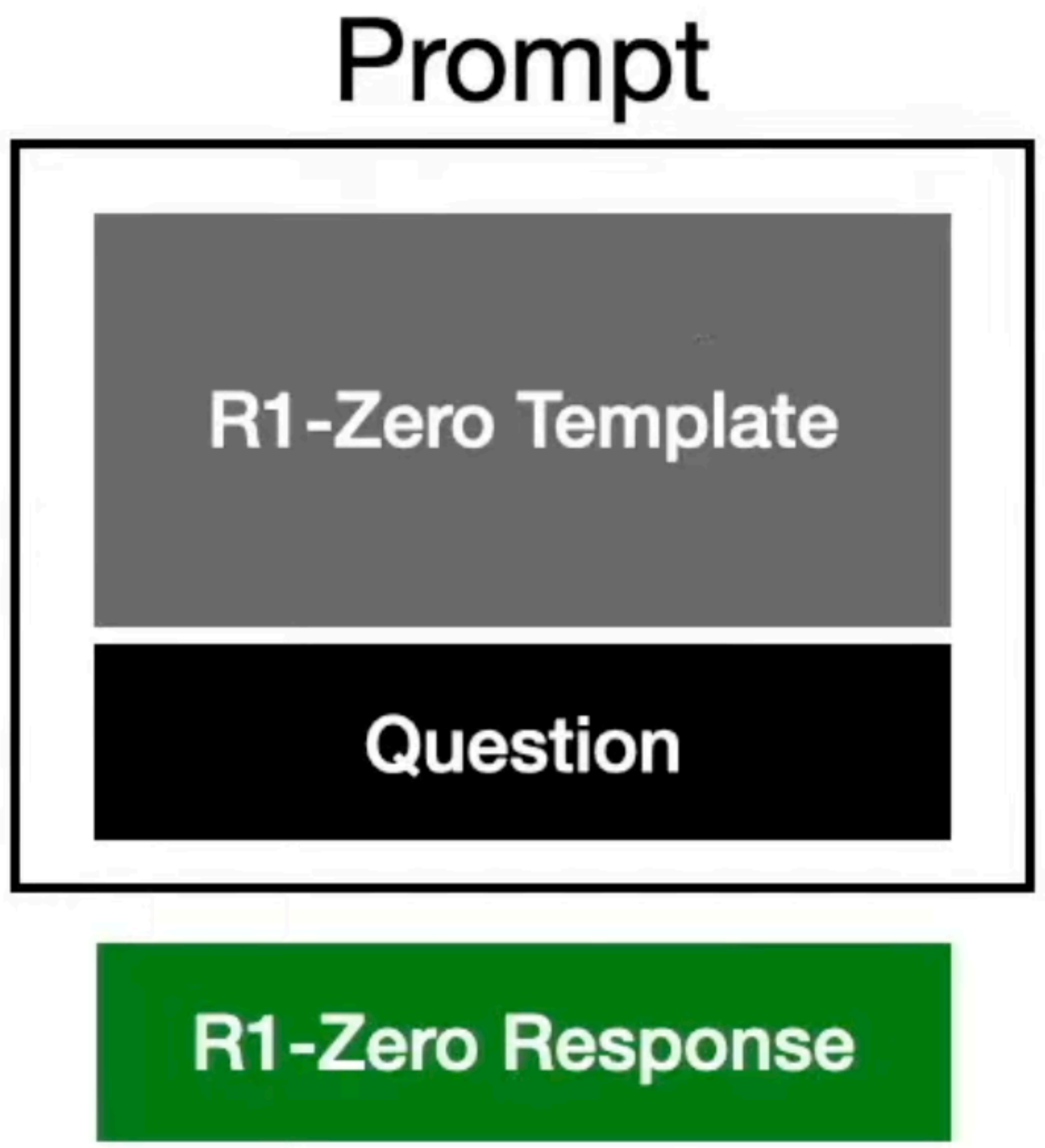
Обучение модели отвечать в нужном формате



Примеры с reasoning



Примеры с chain-of-thought



Примеры из R1-Zero

Step-2: RL как у R1-Zero

Награда за:

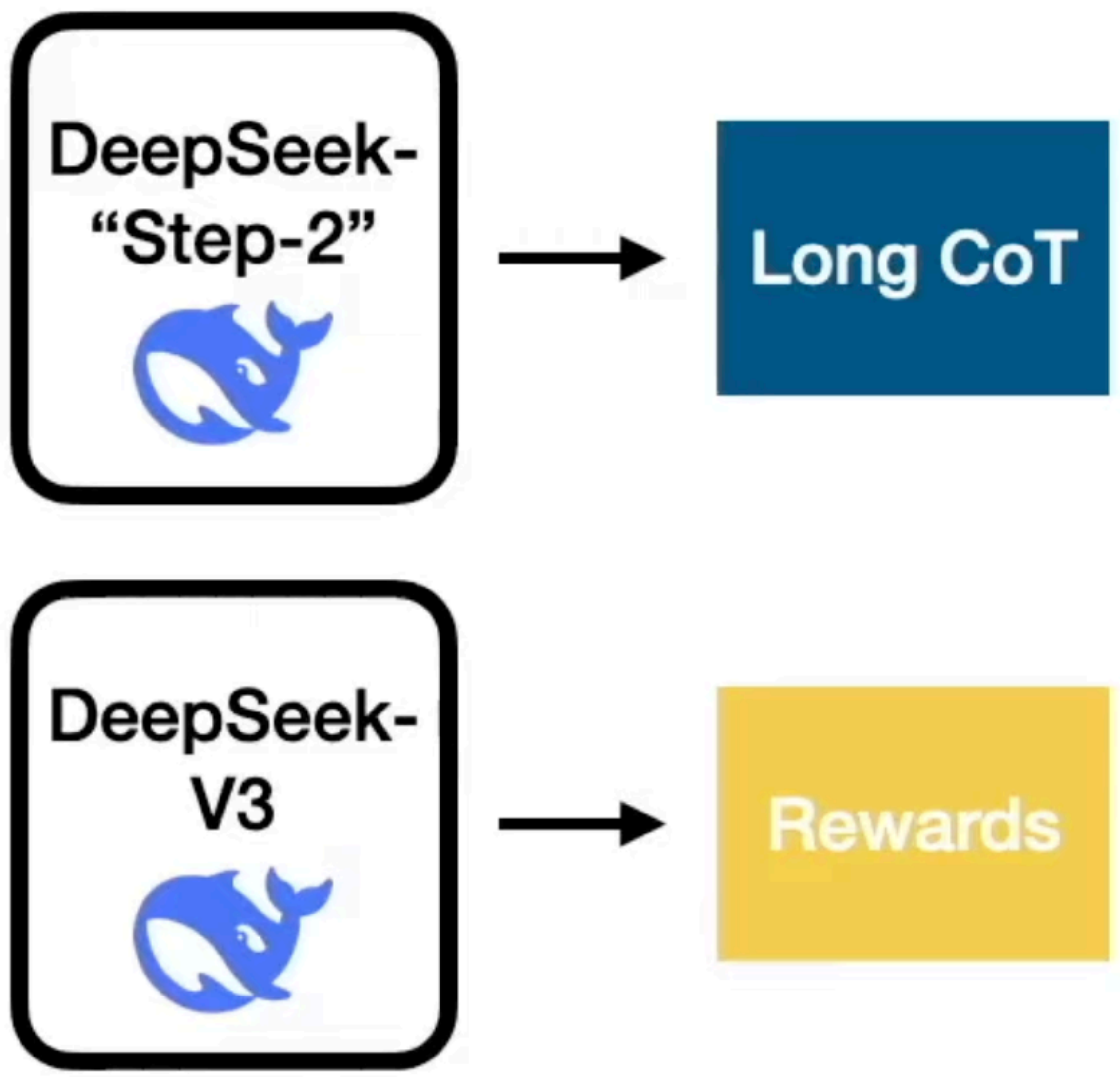
1) Точность ответа

2) Соответствие формату

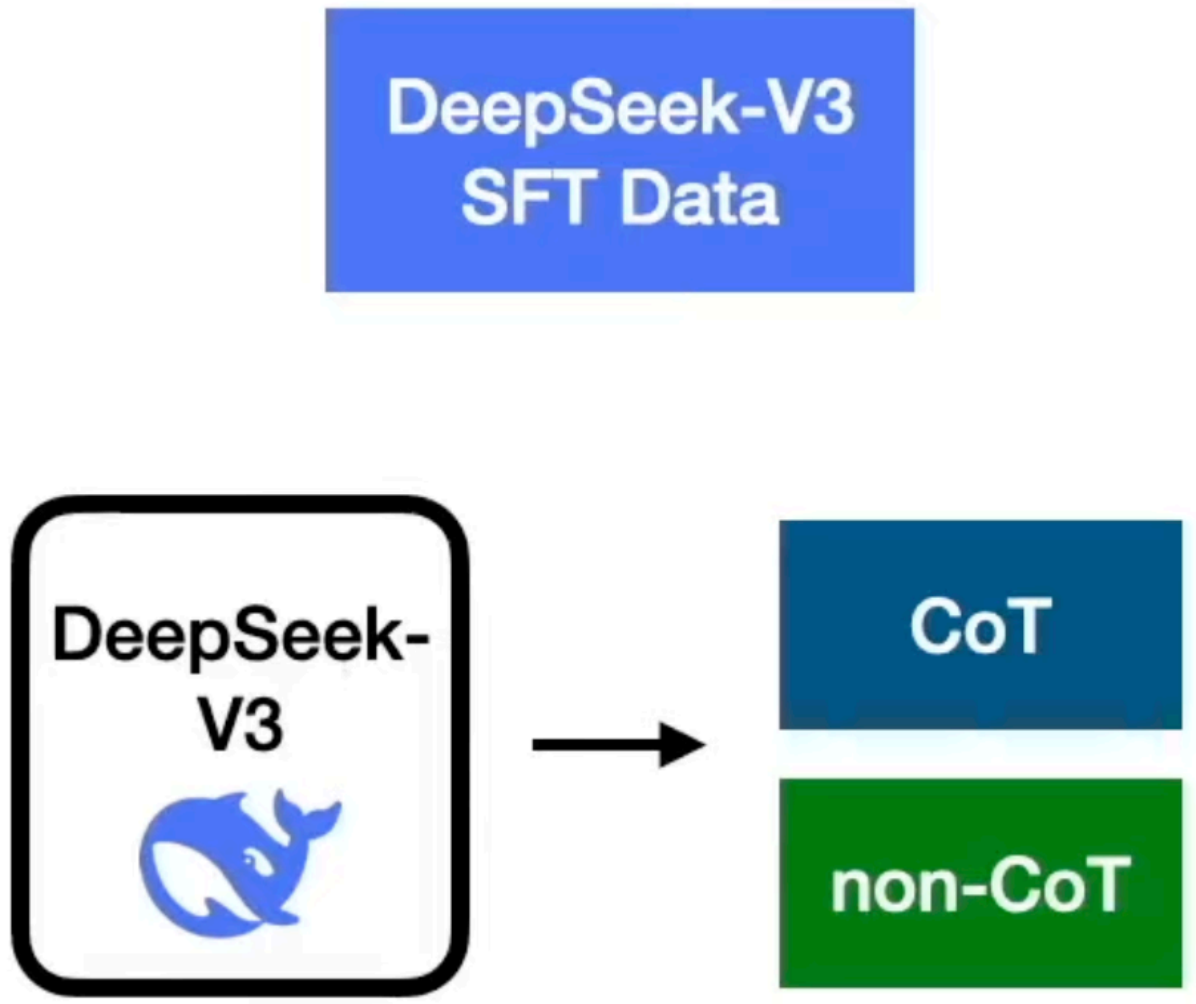
3) Соответствие языку

Step-3: SFT на смеси данных

На данный момент модель не умеет не думать



Reasoning примеры
600k



не Reasoning примеры
200k

Step-4: Alignment (RL + RLHF)

Награда за:

1) Точность ответа

2) Соответствие формату

3) Соответствие языку

R1-Zero-like

+

Награда за:

1) Полезность

2) Безопасность

RLHF

Вместо заключения

На данных момент качество LLM ограничено размером обучающей выборки

RL позволяет снять эти ограничения, так как агент может вечно учиться в среде